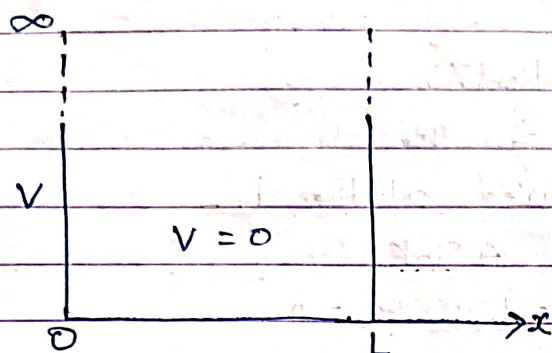
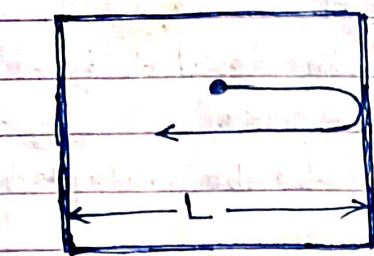


The Particle in a Box: Energy Quantization



Let us consider a particle be moving inside a box along x -direction between $x=0$ and $x=L$. The particle is bouncing back and forth between the walls of the box. The walls are infinitely hard. The particle does not lose energy when it collides with such hard walls, so its total energy E remains constant. The box is supposed to have

walls of infinite height at $x=0$ and $x=L$. The particle has mass m and its position x at any instant is given by $0 < x < L$.

The potential energy V of the particle is infinite on both sides of the box, while V is a constant - say 0 for convenience - on the inside the box.

In terms of the boundary conditions imposed by the problem, the potential function is

$$V = 0 \quad \text{for } 0 < x < L$$

$$V = \infty \quad \text{for } x \leq 0$$

$$V = \infty \quad \text{for } x \geq L$$

Since the particle cannot have infinite amount of energy, it cannot exist outside the box and so its wavefunction ψ is 0 for $x \leq 0$ and $x \geq L$. Now our problem is to find ψ within the box i.e. between $x=0$ and $x=L$.

Within the box Schrödinger's equation becomes

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\psi = 0 \quad (\text{since } V=0) \quad \text{--- (i)}$$

The total derivative $\frac{d^2\psi}{dx^2}$ is used instead of the partial derivative $\frac{\partial^2\psi}{\partial x^2}$ because ψ is a function of x only in this problem.

$$\text{Let } \frac{2mE}{\hbar^2} = k^2$$

The equation (i) reduces to

$$\frac{d^2\psi}{dx^2} + K^2\psi = 0$$

The general solution of this equation is

$$\psi = A \sin Kx + B \cos Kx \quad \text{--- (ii)}$$

where A and B are constants to be evaluated.

Applying boundary condition.

$$\text{when } x=0, \quad \psi = 0$$

Therefore, from equation (ii)

$$0 = B$$

Thus, the required solution is

$$\psi = A \sin Kx$$

$$\text{Now when } x=L, \quad \psi = 0$$

$$\text{i.e. } A \sin KL = 0$$

Since $A \neq 0$, therefore

$$\sin KL = 0$$

$$= \sin n\pi \quad \text{where } n=1, 2, 3, \dots$$

$$KL = n\pi$$

$$\text{or } \sqrt{\frac{2mE}{\hbar^2}} L = n\pi \quad \text{--- (iii)}$$

It is clear that energy of the particle can have only certain values, called the eigenvalues. These eigenvalues, constituting the energy levels of the system, are

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad \text{--- (iv)}$$

where $n=1, 2, 3, \dots$

The integer n corresponding to the energy level E_n is called its quantum number. A particle confined to a box cannot have an arbitrary energy. Inside the box, the particle can have only the discrete energy values given by equation (iv).

It is significant that the electron cannot have zero energy; if it did, the electron wave function ψ would have to be zero everywhere in the box, and this means that the electron cannot be

present there ($\psi = A \sin \sqrt{\frac{2mE}{\hbar^2}} x$, when $E=0$, $\psi=0$). The exclusion of $E=0$ as a possible value for the energy of a trapped electron, like the limitations of E to a discrete set of definite values, is a quantum mechanical result that has no counterpart in classical mechanics, where all energies, including zero, are presumed possible.

[Note:- In the domain of everyday experience, we don't aware of energy quantization. For this let us compute the permitted energy levels of (1) an electron in a box 1 Å wide, and (2) a 10 gram marble in a box 10 cm wide.

In case of 1, we have

$m = 9.1 \times 10^{-31} \text{ kg}$, $L = 1 \text{ Å} = 10^{-10} \text{ m}$, so that the permitted electron energies are

$$E_n = \frac{n^2 \pi^2 \times (1.054 \times 10^{-34} \text{ J-s})^2}{2 \times 9.1 \times 10^{-31} \text{ kg} \times (10^{-10} \text{ m})^2}$$

$$= 6 \times 10^{-18} n^2 \text{ Joule}$$

$$= 38 n^2 \text{ eV.}$$

The minimum energy the electron can have $E_1 = 38 \text{ eV}$. (For $n=1$)

$$E_2 = 4E_1 = 152 \text{ eV}$$

$$E_3 = 9E_1 = 342 \text{ eV}$$

$$E_4 = 16E_1 = 608 \text{ eV. and so on.}$$

These energy levels are sufficiently far apart to make the quantization of electron energy in such a box conspicuous if such a box actually did exist.

In case 2, $m = 10 \text{ gram} = 10^{-2} \text{ kg}$

$$L = 10 \text{ cm} = 10^{-1} \text{ m}$$

the permitted energy levels for marble are

$$E_n = \frac{n^2 \pi^2 \times (1.054 \times 10^{-34} \text{ J-s})^2}{2 \times 10^{-2} \text{ kg} \times (10^{-1} \text{ m})^2}$$

$$= 5.5 \times 10^{-64} n^2 \text{ Joule.}$$

The minimum energy the marble can have is $5.5 \times 10^{-64} \text{ J}$ for $n=1$. A marble with this kinetic energy has a speed of only $3.3 \times 10^{-31} \text{ m/s}$ and is therefore experimentally indistinguishable from a stationary marble. A reasonable

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speed a marble might have, say $1/3 \text{ m/s}$ - which corresponds to the energy level of quantum number $n = 10^{30}$. Here energy permissible energy levels are very close together, so there is no way of determining whether the marble can take on only those energies predicted by equation (iv) or any energy whatever. Therefore, in everyday experience energy seems to be continuous including zero.]

Wave Functions

The wave function of a particle in a box whose energy is E is

$$\psi = A \sin \sqrt{\frac{2mE}{\hbar^2}} x$$

Since possible energies are

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

Substituting E_n for E yields

$$\psi_n = A \sin \frac{n\pi x}{L} \quad \dots (v)$$

for the eigenfunctions corresponding to the energy eigenvalues. Here ψ_n meet all the requirements i.e. for each quantum number n , ψ_n is a single-valued function of x , and ψ_n and $\frac{\partial \psi_n}{\partial x}$ are continuous.

Now it is certain that the particle is somewhere inside the box. Hence for a normalised wave function

$$\int_0^L |\psi_n|^2 dx = 1$$

$$\text{or } A^2 \int_0^L \sin^2 \frac{n\pi x}{L} dx = 1$$

$$\text{or } A^2 \int_0^L \left(\frac{1 - \cos \frac{2n\pi x}{L}}{2} \right) dx = 1$$

$$\text{or } \frac{A^2}{2} \int_0^L dx - \frac{A^2}{2} \int_0^L \cos \frac{2n\pi x}{L} dx = 1$$

$$\text{or } \frac{A^2}{2} L - 0 = 1$$

$$\text{or } A = \sqrt{\frac{2}{L}} \quad \dots (vi)$$

Therefore, normalized wave-function of the particle are

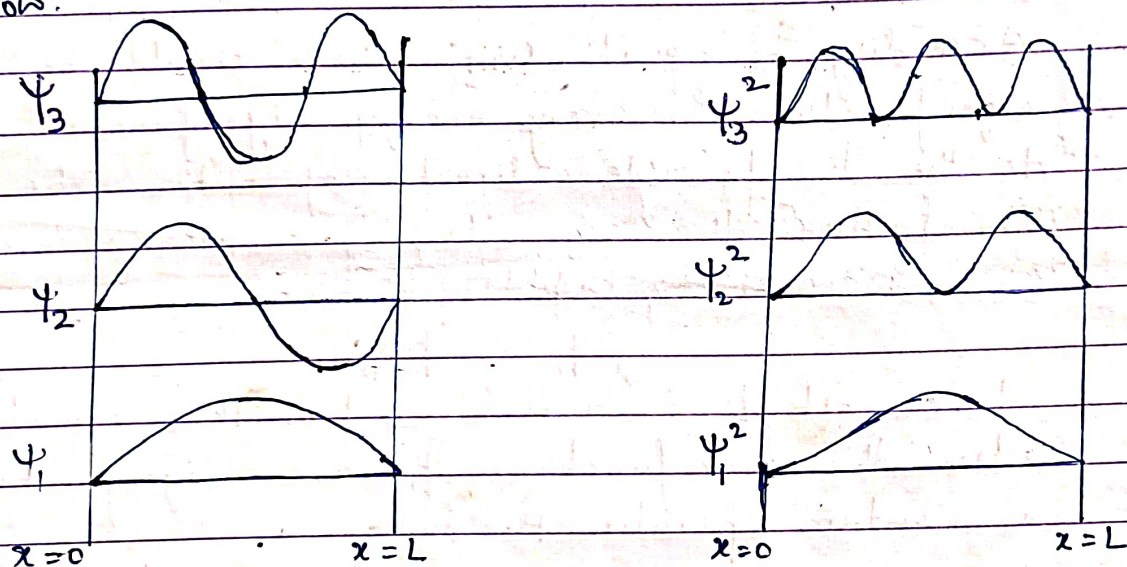
$$\boxed{\Psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}} \quad \text{--- (vii)}$$

Here $\Psi_1 = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}$

$$\Psi_2 = \sqrt{\frac{2}{L}} \sin \frac{2\pi x}{L}$$

$$\Psi_3 = \sqrt{\frac{2}{L}} \sin \frac{3\pi x}{L}$$

The normalized wavefunctions, Ψ_1 , Ψ_2 and Ψ_3 together with the probability densities $|\Psi_1|^2$, $|\Psi_2|^2$ and $|\Psi_3|^2$ are plotted as shown below.



Ψ_n may be negative as well as positive, but $|\Psi_n|^2$ is always positive and since Ψ_n is normalized, its value at a given x is equal to the probability P of finding the particle there. In every case $|\Psi_n|^2 = 0$ at $x=0$ and $x=L$, the boundaries of the box. At a particular point in the box the probability of the particle being present may be very different for different quantum numbers. For instance, $|\Psi_1|^2$ has its maximum value of $\frac{2}{L}$ in the middle of box, while $|\Psi_2|^2 = 0$ there; a particle in the lowest energy level of $n=1$ is most likely to be in the middle of the box, while a particle in the next higher state $n=2$ is never there. Classical physics, however, predicts the same probability for the particle being anywhere in the box.

Momentum Quantization

We know that energy and momentum are related as

$$E = \frac{p^2}{2m}$$

Therefore, $p_n = \pm \sqrt{2mE_n}$

$$= \pm \sqrt{2m} \frac{n\pi\hbar}{\sqrt{2m}L} = \pm \frac{n\pi\hbar}{L} \quad \text{--- (viii)}$$

$\pm$ sign provides the explanation: the particle is moving back and forth and its average momentum for any value of n is

$$p_{av} = \frac{\frac{n\pi\hbar}{L} + (-\frac{n\pi\hbar}{L})}{2} = 0$$

According to equation (viii) there should be two momentum eigenfunctions for every energy eigenfunction, corresponding to the two possible directions of motion. The general procedure for finding the eigenvalues of a quantum-mechanical operator (here $\hat{p}$) is to start from the eigenvalue equation

$$\hat{p}\psi_n = p_n\psi_n \quad \text{--- (ix)}$$

where each p_n is a real number. This equation holds only when the wavefunctions ψ_n are eigenfunctions of the momentum operator $\hat{p}$. Since

$$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$$

the energy eigenfunctions

$$\psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

are not also momentum eigenfunctions, because

$$\frac{\hbar}{i} \frac{d}{dx} \left(\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \right) = \frac{\hbar}{i} \frac{n\pi}{L} \sqrt{\frac{2}{L}} \cos \frac{n\pi x}{L} \neq p_n \psi_n$$

To find the correct momentum eigenfunctions, we use the identity

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} = \frac{1}{2i} e^{i\theta} - \frac{1}{2i} e^{-i\theta}$$

Thus, each energy eigenfunction can be expressed as a linear combination of the two wave functions

$$\psi_n^+ = \frac{1}{2i} \sqrt{\frac{2}{L}} e^{in\pi x/L} \quad \text{--- (x)}$$

$$\psi_n^- = \frac{1}{2i} \sqrt{\frac{2}{L}} e^{-in\pi x/L} \quad \text{--- (xi)}$$

(1.1)

Inserting the first of these wavefunctions in the eigenvalue equation (viii) we have

$$\hat{p} \psi_n^+ = p_n \psi_n^+$$

$$\frac{\hbar}{i} \frac{d}{dx} \psi_n^+ = \frac{\hbar}{i} \frac{1}{\sqrt{2}} \sqrt{\frac{2}{L}} \frac{i n \pi}{L} e^{i n \pi x / L}$$

$$= \frac{n \pi \hbar}{L} \left(\frac{1}{\sqrt{2}} \sqrt{\frac{2}{L}} e^{i n \pi x / L} \right)$$

$$= \frac{n \pi \hbar}{L} \psi_n^+$$

$$= p_n \psi_n^+ \quad \text{where } p_n = \frac{n \pi \hbar}{L}$$

Similarly the wavefunction ψ_n^- yields the momentum eigenfunction

$$p_n^- = -\frac{n \pi \hbar}{L}$$

Therefore ψ_n^+ and ψ_n^- are indeed the momentum eigenfunctions for a particle in a box and equation (viii) correctly states the corresponding momentum eigenvalues.

* Let the speed of the particle is $v = 3 \text{ cm/s}$.

Then $E = \frac{mv^2}{2} = 4.50 \times 10^{-10} \text{ J}$.

Now

$$n = \frac{\sqrt{8mLE}}{h} = \frac{\sqrt{8 \times 10^{-6} \text{ kg} \times 10^{-4} \text{ m}^2 \times 4.5 \times 10^{-10} \text{ J}}}{6.6 \times 10^{-34} \text{ Js}}$$

$$= 9 \times 10^{23}$$

Here the particle of mass is confined in box of length

The value of n is so large that we would never be able to distinguish the quantized nature of the energy levels.

Ques. A particle is known to be in the ground state of an infinite square well with length L . Calculate the probability that this particle will be found in the middle half of the well that is between $\frac{L}{4}$ and $\frac{3L}{4}$.

The probability density is given by $|\psi_n|^2$ with $n=1$ for the ground state. Thus probability is

$$P = \int_{L/4}^{3L/4} |\psi_1|^2 dx = \frac{2}{L} \int_{L/4}^{3L/4} \sin^2 \frac{\pi x}{L} dx$$

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$$= \frac{2}{L} \times \frac{1}{2} \int_{L/4}^{3L/4} 2 \sin^2 \frac{\pi x}{L} dx$$

$$= \frac{1}{L} \left[\int_{L/4}^{3L/4} (1 - \cos \frac{2\pi x}{L}) dx \right]$$

$$= \frac{1}{L} \left[\int_{L/4}^{3L/4} dx - \int_{L/4}^{3L/4} \cos \frac{2\pi x}{L} dx \right]$$

$$= \frac{1}{L} \left\{ \left(\frac{3L}{4} - \frac{L}{4} \right) - \left[\frac{\sin \frac{2\pi x}{L}}{\frac{2\pi}{L}} \right]_{L/4}^{3L/4} \right\}$$

$$= \frac{1}{L} \left(\frac{L}{2} \right) - \frac{1}{L} \frac{L}{2\pi} \left(\sin \frac{2\pi}{L} \frac{3L}{4} - \sin \frac{2\pi}{L} \frac{L}{4} \right)$$

$$= \frac{1}{2} - \frac{1}{2\pi} \left(\sin \frac{3\pi}{2} - \sin \frac{\pi}{2} \right)$$

$$= \frac{1}{2} - \frac{1}{2\pi} (-1 - 1) = \frac{1}{2} + \frac{1}{\pi} = 0.818.$$

Blood urea

Cr. Creatinine

WBC and

BT, CT